

CHEMICAL BONDING

Electronegativity

Electronegativity can be defined as

“The tendency of an atom to attract the shared paired electron or bond forming electron pair toward itself is called electronegativity.”

For example

Cl attract the bond pair toward itself more strongly because Cl has high electronegative then H.



Factor affecting the electronegativity

Following factors affect the value of electronegativity.

- ✓ Size of atom
- ✓ Nuclear charge
- ✓ Shielding effect

1. Size of atom:

When the size of an atom is decreases the distance of nucleus form electron is decrease and the force of attraction of nucleus on the valance electrons is increased. So, more electrons are attracted toward the atom which cause an increase in the value of electronegativity.

2. Nuclear charge:

In period form left to right the numbers of protons are increased which increase the effective nuclear charge. The more positive charge attracts the bonded electrons more powerfully. So, the electronegativity is increased.

3.Shielding effect:

In period form left to right, shielding effect remains constant. But in group from top to bottom shielding effect is increased due to the addition of new shell. The attraction of nucleus on the bonded electrons is decreased which result in decrease of the electronegativity value.

Periodic trend of electronegativity

Trend in period:

In period form left right of the periodic table the nuclear charge is increased but the shielding effect remains constant. Due to the increased nuclear charge the distance of between the nucleus and valence shell is decreased which increase the force attraction of nucleus on the bonded electrons. So, form left to right electronegativity is

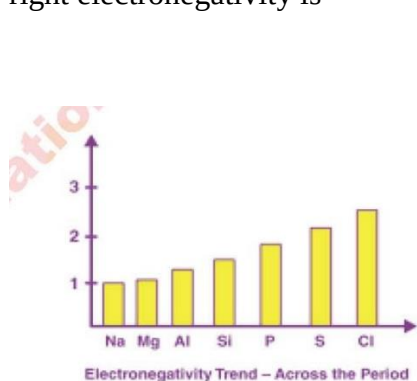


Figure 3.2: Trend of Electronegativity of third period elements.

1 H 2.20																	5 B 2.04	6 C 2.55	7 N 3.04	8 O 3.44	9 F 3.98	
3 Li 0.98	4 Be 1.57	Pauling Electronegativity Values																13 Al 1.61	14 Si 1.90	15 P 2.19	16 S 2.58	17 Cl 3.16
11 Na 0.93	12 Mg 1.31	19 K 0.82	20 Ca 1.00	21 Sc 1.36	22 Ti 1.54	23 V 1.63	24 Cr 1.66	25 Mn 1.55	26 Fe 1.83	27 Co 1.88	28 Ni 1.91	29 Cu 1.90	30 Zn 1.65	31 Ga 1.81	32 Ge 2.01	33 As 2.18	34 Se 2.55	35 Br 2.96				
37 Rb 0.82	38 Sr 0.95	39 Y 1.22	40 Zr 1.33	41 Nb 1.6	42 Mo 2.16	43 Tc 1.9	44 Ru 2.2	45 Rh 2.28	46 Pd 2.20	47 Ag 1.93	48 Cd 1.69	49 In 1.78	50 Sn 1.96	51 Sb 2.05	52 Te 2.1	53 I 2.66						
55 Cs 0.79	56 Ba 0.89	57 La 1.1	72 Hf 1.3	73 Ta 1.5	74 W 2.36	75 Re 1.9	76 Os 2.2	77 Ir 2.20	78 Pt 2.28	79 Au 2.54	80 Hg 2.00	81 Tl 1.62	82 Pb 2.33	83 Bi 2.02	84 Po 2.0	85 At 2.2						
87 Fr 0.7	88 Ra 0.9																					

Nature of bond in term of Pauling electronegativity value

Ch#3: Chemical Bonding**By: Sir Amjad (Lecturer Chemistry)**

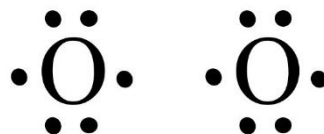
From Ionic bond Difference of electronegativity between two atoms = higher than 1.8

For polar covalent bond difference = 1.8 to 0.4

For Non-polar covalent bond difference = 0.4 or low

Examples:**1- Oxygen molecule (O₂)**

Both oxygen atom has electronegativity 3.44. The difference of electronegativity is zero. So, none of the oxygen atom pulls the bond electrons toward itself more powerfully. So, it has non-polar covalent bond.

**2. Potassium Chloride (KCl):**

Electronegativity of K = 0.82

Electronegativity of Cl = 3.16

Difference = 3.16 - 0.82 = 2.34

As the difference is greater than 1.8, so bond between K and Cl is ionic

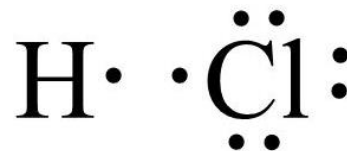
**3. Hydrogen Chloride (HCl)**

Electronegativity of H = 2.20

Electronegativity of Cl = 3.16

Difference = 3.16 - 2.20 = 0.96

So, the bond will be polar. As the Cl has high electronegativity than H so, it pulls the electrons pair toward itself more strongly.

**Covalent Character**

Covalent character depends upon the polarizing power of cation which in turn depends upon the oxidation power of cation.

In period from left to right polarizing power of cation increases so it increases the covalent character.

Example:

AlCl₃ has more covalent character than MgCl₂. It is because Al can lose three electrons to form Al³⁺ cation while Mg can lose two electrons to form Mg²⁺ cation. As polarizing power of Al cation is higher than Mg. So, AlCl₃ is more covalent.

Shapes of Molecules**Valence shell Electron Pair Repulsion (VSEPR) theory**

Suggested by Sidwick and Powell in 1940.

Later, explained by Nyholm and Gillespie in 1957

Purpose:

This theory is used to explain shapes and geometry of molecules and ions (for non-transition elements)

Basic Assumption:

"Electron pairs are arranged around the central atom to remain at maximum distance apart to keep repulsion minimum."

Following types of molecules are studied

AX₂, AX₃, AX₄, AX₅, AX₆

A = A central atom (polyvalent atom)

X = side atom (H, X, O)

2, 3, 4, 5, 6 are numbers of electron pairs (Bonding electron pair, Non-bonding electron pair or lone pair)

Postulates of VSEPR theory:

1. The postulates of the VSEPR theory are listed below.
2. In polyatomic molecules (i.e., molecules made up of three or more atoms), one of the constituent atoms is identified as the central atom to which all other atoms belonging to the molecule are linked.
3. The total number of valence shell electron pairs decides the shape and angle of the molecule.
4. The electron pairs tend to orient themselves in a way that minimizes the electron-electron repulsion between them and maximizes the distance between them.
5. When the central atom of the molecule is surrounded by bond pairs of electrons, then the asymmetrically shaped molecule can be expected.
6. When the central atom is surrounded by both lone pairs and bond pairs of electrons, the molecule would tend to have a distorted shape.
7. The strength of the repulsion is strongest in two lone pairs and weakest in two bond pairs. The order of repulsion is
8. Lone pair-lone pair > lone pair-bond pair > bond pair-pair

1. Shapes of molecules containing two electron pairs

AX₂ types molecules:

- ✓ Central atom is surrounded by two bond pairs.
- ✓ No lone
- ✓ Shape is linear.
- ✓ Bond angle is 180°.
- ✓ Examples
- ✓ Halides of group (IIA = Be, Mg, Ca, Sr, Ba) elements.
- ✓ Like BeCl₂, BeF₂, MgCl₂, MgF₂, CaCl₂, HgCl₂
- ✓ CO₂
- ✓ CS₂
- ✓ C₂H₂

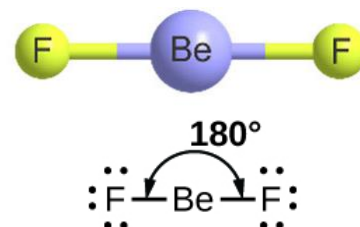
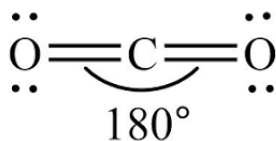
Structure of BeF₂

Two electron pairs around the central Be.

Structure is linear.

Bond angle is 180°.

Structure of CO₂:



2. Shapes of Molecules containing three Electron pairs:

A) Three bonding pairs and no lone pair (AX₃ type molecules)

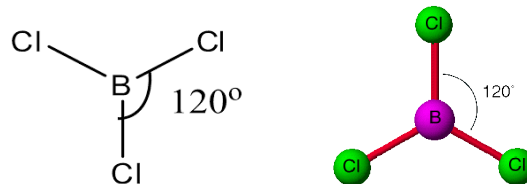
- ✓ Angle 120
- ✓ Shape triangular planer or trigonal
- ✓ (IIIA group = B, Ga, In, Tl) Hydride and halides
- ✓ Hydride = BH₃, AlH₃, GaH₃, InH₃)
- ✓ Halides = BCl₃, AlCl₃, GaCl₃, InCl₃
- ✓ BF₃, AlF₃, GaF₃, InF₃
- ✓ Ions. CO₃²⁻, NO₃⁻, C₂H₄

Ch#3: Chemical Bonding

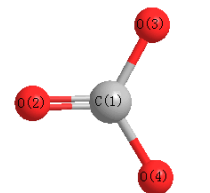
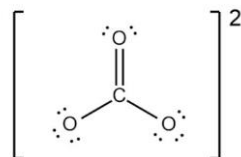
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Examples:**BCl₃**

- ✓ Central atom B has three valence electrons.
- ✓ Three bond electron pair around central atom.
- ✓ Angle is 120° to minimize repulsion.
- ✓ Geometry: trigonal planer

**CO₃²⁻**

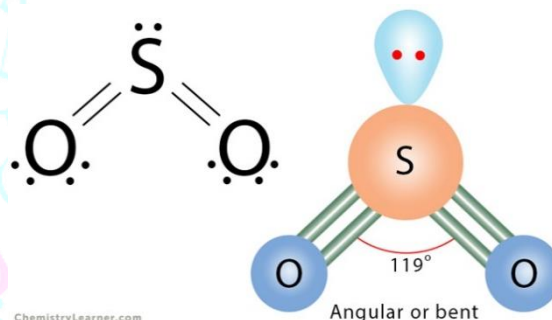
- ✓ Central atom C has four valence electrons.
- ✓ Each O has six valence electrons.
- ✓ Three bond pairs around central C.
- ✓ Bond angle :120°
- ✓ Geometry: Trigonal planer

**B) Two bond pairs and one lone pair (AX₂E type molecules):**

- ✓ (Sn, Pb=IVA group) halides (SnCl₂, PbCl₂) and SO₂, SO₃
- ✓ Electron pair = 3(2 bp + 1lp)
- ✓ Angle less than 120° due to repulsion.
- ✓ Structure = distorted (bent, angular or V-shaped)

Example:**Sulfur dioxide (SO₂)**

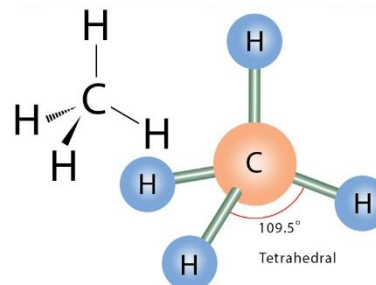
- ✓ Central atom S has six valence electrons.
- ✓ No of electrons pair around central atom: Three
- ✓ Number of bond pairs: two
- ✓ Number of lone pairs: One
- ✓ BP-BP interactions= One
- ✓ BP-LP interactions= Two
- ✓ Angle less than 120°
- ✓ Shape= Bent or V-shaped

**3. Shapes of Molecules containing four electrons pairs:****A) Four bond pairs and no lone pair (AX₄ type molecules)**

- ✓ IVA=C, Si, Ge Sn, Pb
- ✓ Hydride of IVA = CH₄, GeH₄, SiH₄,
- ✓ Halides of IVA= CCl₄, SiCl₄, SnCl₄, PbCl₄
- ✓ Ions= SO₄²⁻, CrO₄²⁻, NH₄⁺, PH₄¹⁺

Example (CH₄):

- ✓ Central atom: C
- ✓ Electron pairs: Four
- ✓ Bond pair: four
- ✓ Lone pair: No
- ✓ Bond angle :109.5
- ✓ Geometry: Tetrahedral

**B) Three bond pairs and one lone pair (AX₃E type molecules):**

- ✓ Central atom is surrounded by four electron pairs
- ✓ Three bond pairs and one lone pair.
- ✓ Angle =107.5
- ✓ Shape: Trigonal pyramid

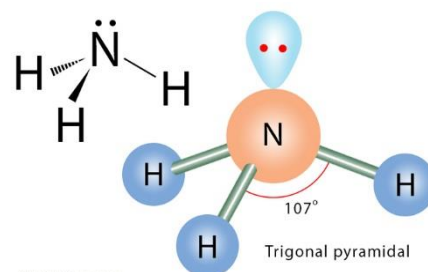
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- ✓ VA group =(N, P, As, Sb ,Bi)
- ✓ Hydrides=(NH₃, PH₃, AsH₃, SbH₃,)
- ✓ Halides=(NF₃, NCl₃, PCl₃, PF₃, PBr₃)

Example: NH₃

- ✓ Central atom: N
- ✓ Number of valence electrons in N: Five
- ✓ Number of electron pairs: Four
- ✓ Number of bond pairs; Three
- ✓ Number of lone pairs: one
- ✓ Bond angle: less than 109.5 (lp-Bp repulsion)

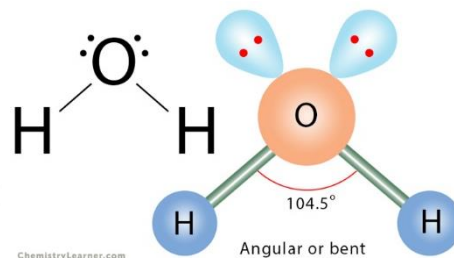


C) Two Bond Pairs and Two Lone Pairs: (AX₂E₂ type molecules)

- ✓ Central atom is surrounded by four electrons pairs.
- ✓ Number of bond pair =two
- ✓ Number of lone pairs =two
- ✓ Shape bent or V-shaped
- ✓ Three type of repulsion = ln-ln>ln-bp>bp-bp
- ✓ Angle = Deviation form tetrahedral bond angle
- ✓ VIA group = (O, Si, Se,Te, Po)
- ✓ Hydride= H₂O, H₂Si, H₂Se, H₂Te

Example (H₂O)

- ❖ Central atom :O
- ❖ Number of valence electrons: Six
- ❖ Number of Electron pairs: Four
- ❖ Number of bond pairs: two
- ❖ Number of lone pairs: two
- ❖ Bond angle: 104.5
- ❖ Geometry: Angular, bent or V-shaped.



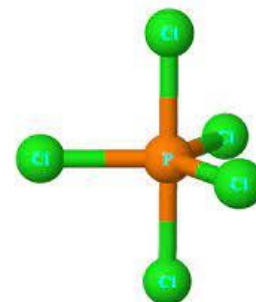
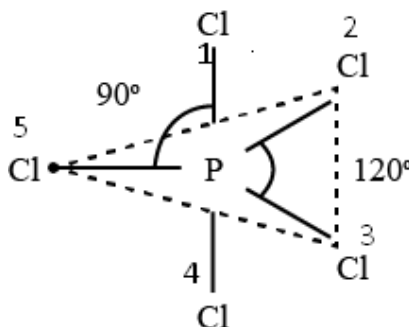
Note: There is significant decrease in the bond angle due to lone Lp-Lp, Lp,Bp and Bp-Bp repulsion. The bond angle is even less than NH₃ due to the presence of two lone pairs.

4. Shapes of molecules containing Five electrons pairs: (AX₅ type)

Some atom doesn't follow the noble gas electronic configuration and have expanded octet state.

Example (PCl₅)

- ✓ Central atom: P
- ✓ Number of electrons pairs: Five
- ✓ Number of bond pairs: Five
- ✓ Number of lone pairs: No
- ✓ Three Cl lies in same plane as P:
- ✓ Bond angle Cl-P-Cl: 120
- ✓ Two Cl lie above and below this plane
- ✓ Bond angle Cl-Cl: 90
- ✓ Geometry: Trigonal bipyramidal



5. Shapes of molecules containing Six electron pairs: (AX₆ type molecules)

- ✓ Example: SF₆, SeF₆, TeF₆
- ✓ SF₆ molecule:
- ✓ Central atom: S
- ✓ Number of electron pairs: six
- ✓ Number of Bond pairs: Six
- ✓ Number of lone pairs: No
- ✓ Expanded octet state
- ✓ **Geometry:** Octahedral
- ✓ **Regular structure:**
- ✓ All side atom are same
- ✓ F-S-F bond angle :90

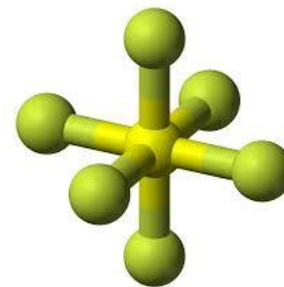
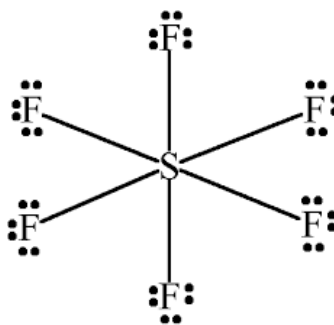


Table 3.4: Shapes of Molecules

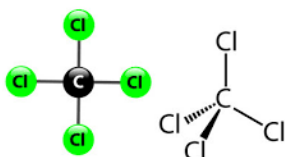
Summary of Shapes of Molecules						
Number of valence electrons	Number of bonding pairs of electrons	Number of lone pairs of electrons	Total number of electron pairs	Arrangement of orbitals	Arrangement of atoms / Angle	Examples
4	2	0	2	Linear	Linear 180°	BeCl ₂ , HCN, CO ₂
6	3	0	3	Trigonal planar	Trigonal planar 120°	BF ₃ , AlCl ₃ , NO ₃ ¹⁻
8	2	1	3	Tetrahedral	Bent 118°	SnCl ₂
	4	0	4		Tetrahedral 109.5°	CH ₄ , SiCl ₄ , SO ₄ ²⁻ , NH ₄ ¹⁺
	3	1	4		Trigonal Pyramid 107°	NH ₃ , PCl ₃ , ClO ₃ ¹⁻
10	2	2	4	Trigonal bipyramid	Bent 104.5°	H ₂ O, H ₂ S
	5	0	5		Trigonal bipyramid 90° and 120°	PCl ₅ , PI ₅
12	6	0	6	Octahedral	Octahedral 90°	SF ₆

Concept Assessment 3.2

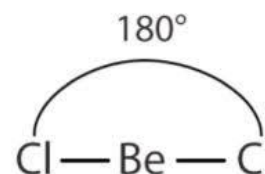
1. Predict the shape of following molecules

A) Tetra chloromethane (CCl₄)

Tetrahedral shape

b) Beryllium chloride (BeCl₂)

Linear shape

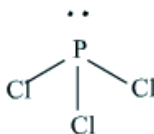


Ch#3: Chemical Bonding

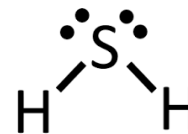
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C) Phosphorus (III) chloride

Trigonal pyramidal

**2. Draw dot and cross structure for following molecules and then predict their shapes.**i) Hydrogen sulfide, H_2S

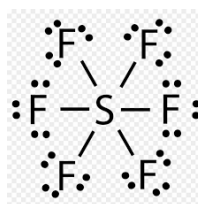
Bent or V shaped

**2. Draw dot and cross structure for following molecules and then predict their shapes.**i) Hydrogen sulfide, H_2S

Bent or V shaped

a) Dot and Cross diagram for SeF_6 B) Shape of SeF_6 :

Octahedral

**C) Shape of phosphorus pentachloride (PCl_5)**

Trigonal bipyramidal

**Valence Bond Theory**

Proposed by Heitler and London (1927)

Developed by Pauling and Slater

According to Valence Bond Theory,

"A Bond is formed by the by the overlapping of partially filled atomic orbital of one atom to the partially filled atomic orbital of other atom and overlapping orbitals retain their identity."

Main point of VBT:

- ✓ A bond between two is formed by the overlapping of partially filled atomic orbitals of two atoms.
- ✓ Atomic orbitals retain their identity after overlapping.
- ✓ Number of bond formed are equal to the number of unpaired electrons in valence shell of atom.
- ✓ Overlapping orbitals must have same symmetry and same sign for bond formation.
- ✓ By head to head overlapping to two atomic orbitals, a single bond is formed which is called sigma bond.
- ✓ By sideways overlapping of additional atomic orbitals multiple bond are formed.
- ✓ Electron pair stabilized after overlapping
- ✓ Energy is released by overlapping of orbitals.
- ✓ Greater the overlap between atomic orbitals, greater energy is released and strong will be the bond.

Overlapping:

The term overlap means two atomic orbitals share common region in space.

Overlapping leads to bond formation and energy is released. (exothermic)

i) Nature of orbitals:

S-orbital ---spherically symmetrical and non-directional =>less overlapping

P,d,f orbitals --(directional) more overlapping

ii) Nature of overlapping

Head to head --sigma bond--more overlapping---strong bond

Side way ----pi bond ---less overlapping ---weak bonding

iii) Bond strength

Bond strength is inversely proportional to value of n

$1-1 > 2-2 > 3-3 > 4-4$

$1s-1s > 2s-2s > 3s-3s >$

$2s-2p > 3s-3p >$

Types of Overlapping and Nature of covalent bond:

Two types of covalent bonds are obtained by overlapping.

i) Sigma Bond (σ)

A sigma bond is formed by the head on overlapping of atomic orbitals. A single bond is always a sigma bond.

For example

Overlapping of s-s, s- p_x , s- p_y , s- p_z , p_x - p_x orbitals.

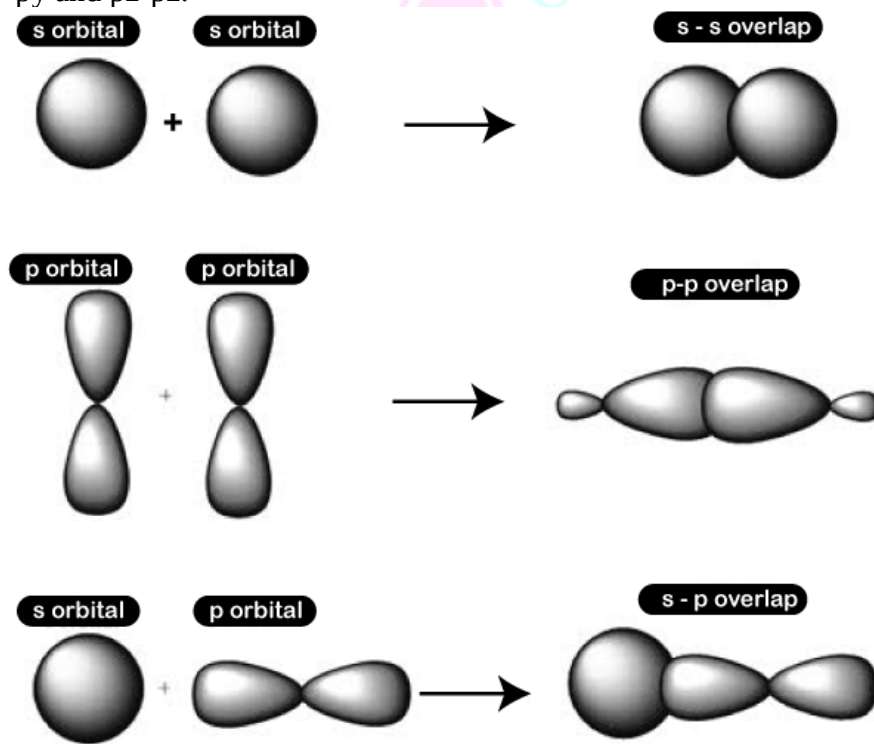
ii) Pi bond (π)

A pi bond is formed by the sideways overlapping of p orbitals.

In double bond 2nd bond is always a pi bond and in triple bond 2nd and 3rd both bonds are pi bonds.

For example:

Overlapping of p_y - p_y and p_z - p_z .



Different between sigma (σ) and pi(π) Bond

sigma (σ) Bond	pi(π) Bind
It is formed by end to end overlapping of atomic orbitals	It is formed by side way or parallel overlap of orbitals
The orbitals involved are s-s,s-p or p-p	These bonds are formed by overlapping of p-p orbitals only.
It is strong bond	It is a weak bond
Free rotation about the sigma bond is possible.	Rotation is restricted in case of pi bond.

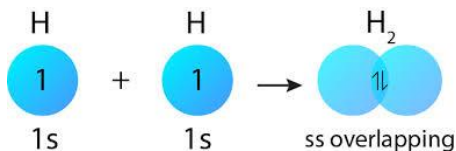
Applications of Valence Bond Theory

1. Single Bond Formation:

A) Formation of H₂ molecules:

- ✓ Electronic configuration of H=1s¹
- ✓ Each half-filled orbital of H overlap to form H-H bond
- ✓ The electron density in between the line joining the two nuclei.

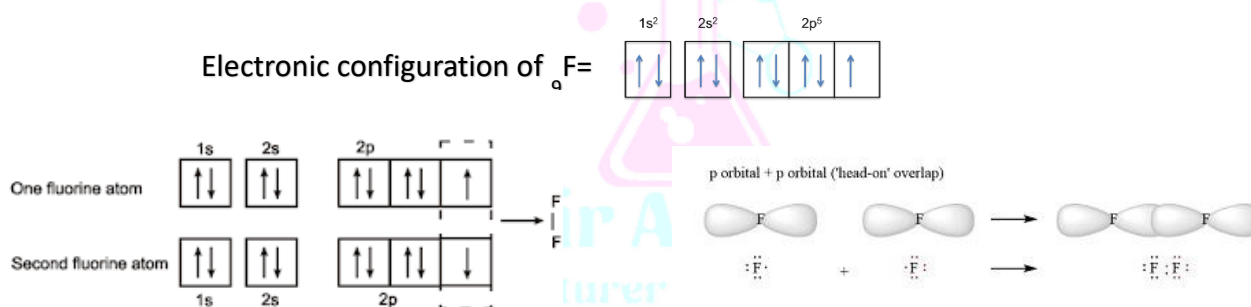
VBT diagram



B) Formation of F₂ molecule:

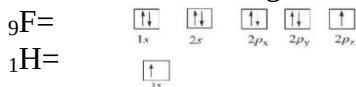
F has one half-filled atomic orbital.

Half-filled 2p_z atomic orbitals of each fluorine head on overlap to form a sigma bond.



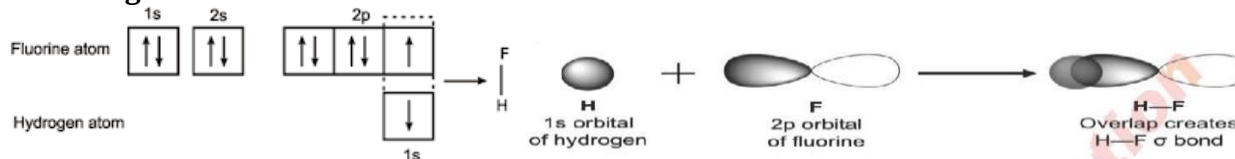
C) Formation of HF:

The electronic configuration of H and F



Half-filled 1s orbital of H overlap with half-filled 2p_z orbital of F to form a sigma bond between H and F

VBT diagram of HF:



2. Double Bond Formation:

Formation of O₂ molecule:

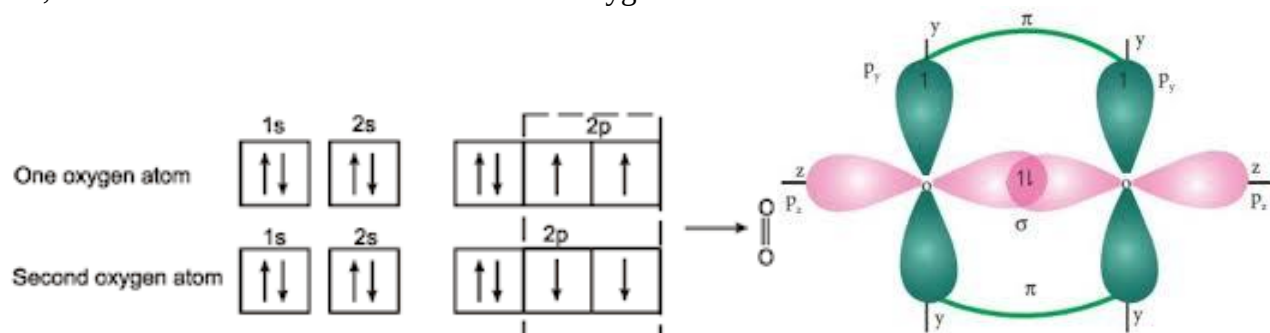
The electronic configuration of oxygen is

So, two half-filled 2p_y and 2p_z orbitals are present.

One half filled 2p orbital of 1st oxygen atom head on overlap with half-filled 2p orbital of other oxygen atom to form a sigma bond.

2nd half-filled 2p orbital also parallel overlap with 2p half-filled orbital of other oxygen to form a pi bond.

So, double bond is formed between the two oxygen atoms.

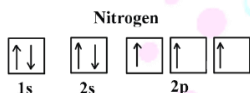


3. Triple Bond formation:

Formation of N₂ molecule

N₂ molecule has two N atom.

The electronic configuration is

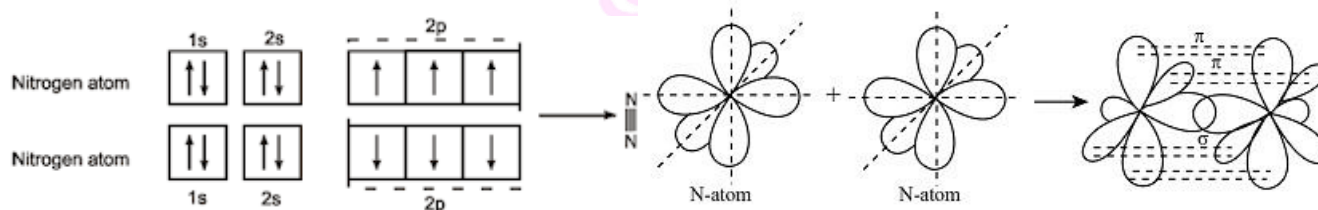


Each N has three half filled p orbitals.

Half filled 2p_x orbitals head on overlap to form a sigma bond.

Half filled 2p_y and 2p_z orbitals parallel overlap to form two pi bonds.

Triple bond is formed between two nitrogen atoms.



Limitation of VBT

VBT fails to explain;

- ✓ Paramagnetic behavior of O₂
- ✓ Bond angle in H₂O and NH₃
- ✓ Equivalent tetra valency of carbon.

Atomic Orbital Hybridization

Hybridization can be defined as;

"The process in which mixing of atomic orbitals having different shapes and energy to form hybrid orbitals with same shape and energy but different orientation is called hybridization."

Explanation:

During hybridization electron in ground state is promoted to excited state to increase the unpaired electron.

The energy required for the excitation is compensated by the energy released during bond formation.

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As VBT cannot explain the tetra-valency of C atom. Hybridization can well explain the equivalent tetra-valency of carbon.

Types of Hybridization:

Based on the nature and number of atomic orbitals participating in intermixing there are following types of hybridization.

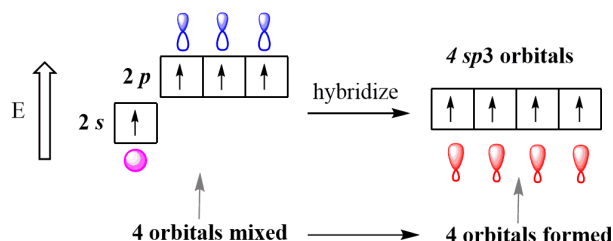
sp^3 , sp^2 , sp , dsp^3 , d^2sp^2 etc

 sp^3 hybridization:

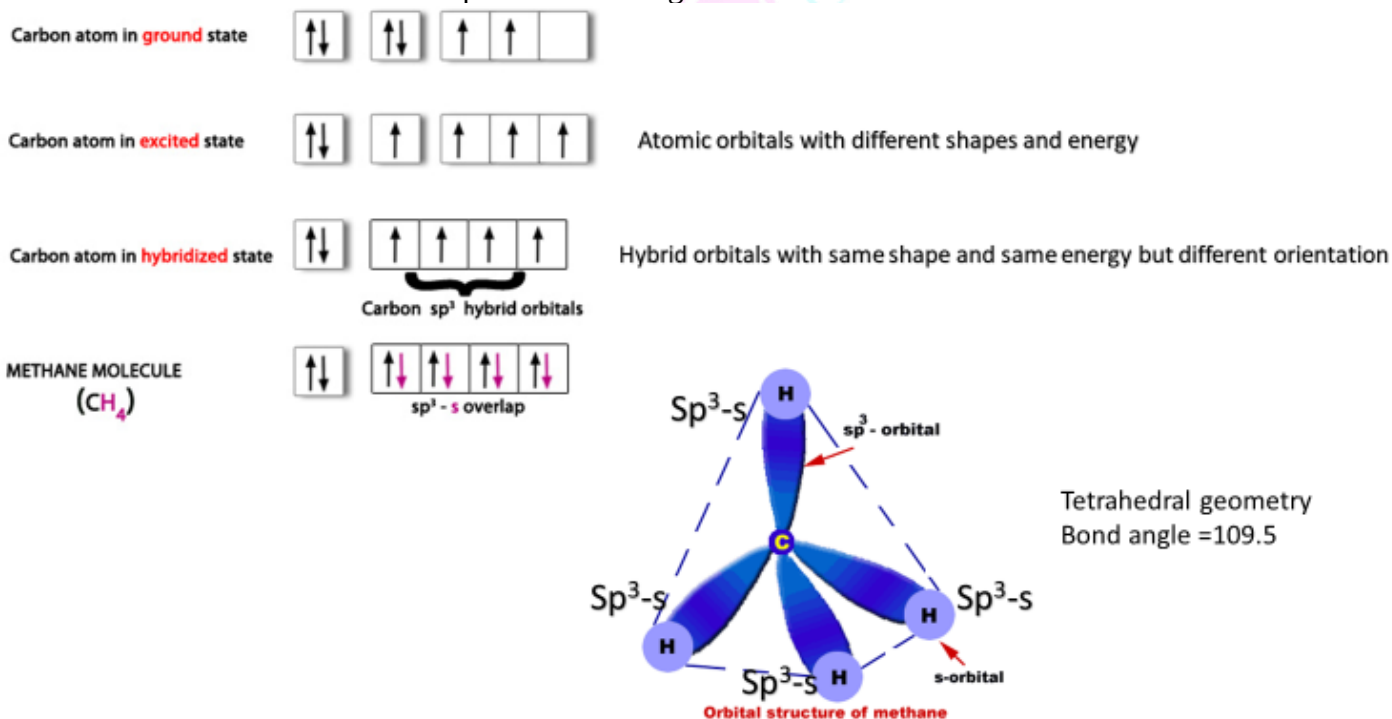
"In this hybridization one 's' and three 'p' atomic orbitals (slightly different energy) to form four sp^3 equivalent hybrid orbitals."

Sp^3 hybrid orbitals have 25% s and 75% p character

Examples CH_4 , C_2H_6 , NH_3 , H_2O

**1. Methane (CH_4)**

- ✓ In the formation of CH_4 molecule, in excited state one 2s and three 2p atomic orbitals of carbon intermix to form four sp^3 half-filled hybrid orbitals having same shape and energy.
- ✓ Three hybrid orbitals are orientated in space around the central atom to form the tetrahedral geometry.
- ✓ These sp^3 hybrid orbitals form sigma bond with 1s half-filled atomic orbital of hydrogen atom.
- ✓ Methane is tetrahedral in shape with bond angle 109.5° .

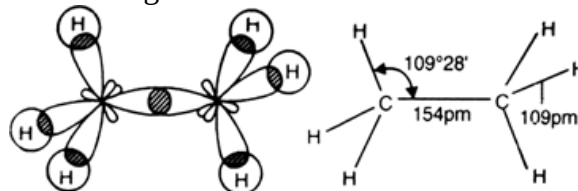
**2. Ethane (C_2H_6):**

- ✓ Each C atom in excited state undergo sp^3 hybridization to form four sp^3 hybrid orbitals.
- ✓ A sigma bond is formed by the sp^3-sp^3 overlap between the two carbon atoms.

Ch#3: Chemical Bonding

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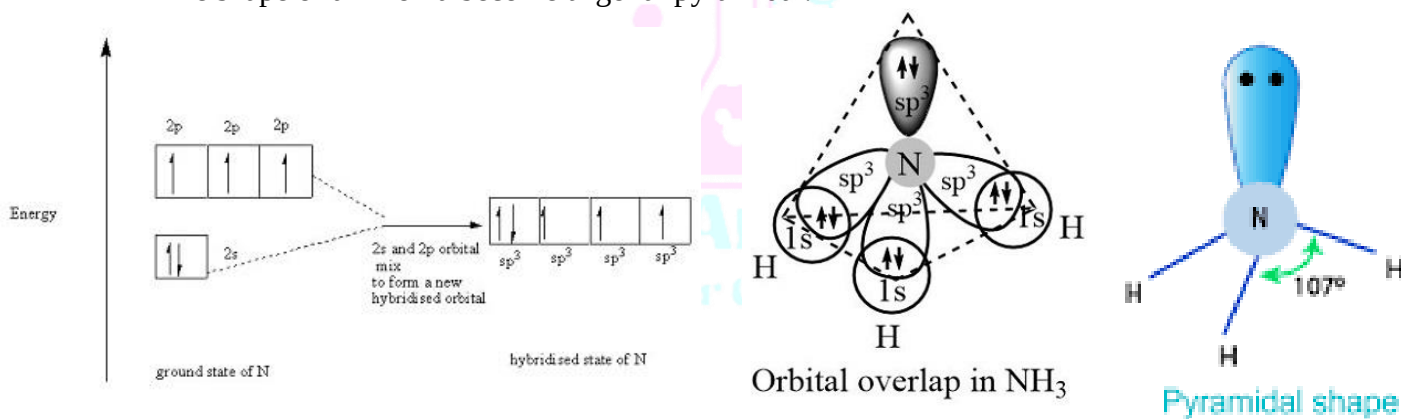
- ✓ Remaining three sp^3 hybrid orbitals of each carbon atom for sigma bond with 1s atomic orbital of hydrogen atom by sp^3 -s overlap.
- ✓ Tetrahedral geometry with bond angle 109.5° is formed.



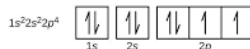
Molecular orbital picture of ethane

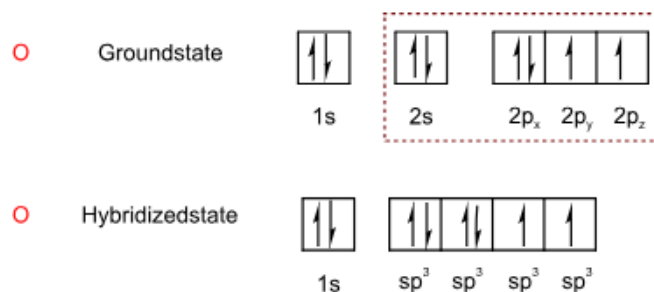
3. Ammonia (NH_3)

- ✓ The electronic configuration of Nitrogen is: $1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$
- ✓ N has three unpaired electrons so, it can form three bonds.
- ✓ Three is also a lone pair electron.
- ✓ One 2s and three 2p atomic orbitals of N intermix to form four sp^3 hybrid orbitals.
- ✓ These hybrid orbitals are orientated tetrahedral symmetry around the central atom.
- ✓ This arrangement give more stability to molecule due to minimization of repulsion.
- ✓ Three out four sp^3 hybrid orbitals form sigma bond with 1s half filled atomic orbital of hydrogen atom by sp^3 -s overlap.
- ✓ One sp^3 completely filled hybrid orbitals remains in the form of lone pair.
- ✓ This lone pair occupy more space than tetrahedral bond angle.
- ✓ So, bond angle decreased to 107° due to the repulsion caused by lone pair over bond pair.
- ✓ The shape of ammonia become trigonal pyramidal.

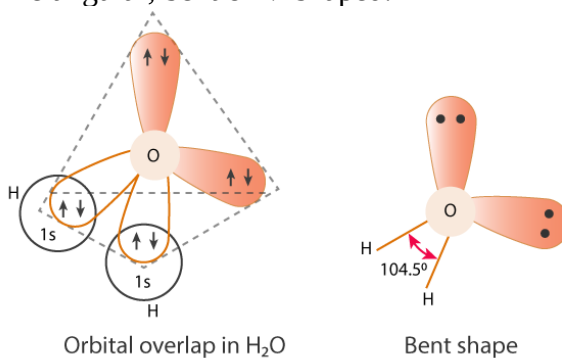
4. Water (H_2O):

- ✓ The electronic configuration of O is:
 - has two unpaired electrons so, it can form two bonds.
- ✓ There are two completely filled atomic orbitals which are lone pair of electrons.
- ✓ Due to two lone pair the bond angle further decreased to 104.5° .
- ✓ One 2s and three 2p atomic orbitals of O intermix to form four sp^3 hybrid orbitals.
- ✓ These hybrid orbitals are orientated tetrahedral symmetry around the central atom.



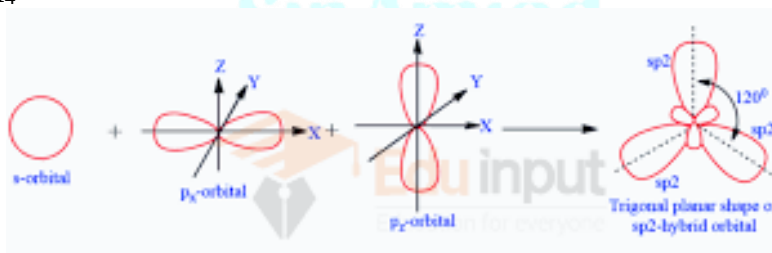


- ✓ Two out of four sp^3 hybrid orbitals form sigma bond with 1s half filled atomic orbital of hydrogen atom by sp^3 -s overlap.
- ✓ Two sp^3 completely filled hybrid orbitals remain in the form of lone pair.
- ✓ This lone pair occupies more space than tetrahedral bond angle.
- ✓ So, bond angle decreased to 104.5° due to the repulsion caused by lone pair over bond pair.
- ✓ The shape of ammonia becomes angular, bent or V-shaped.



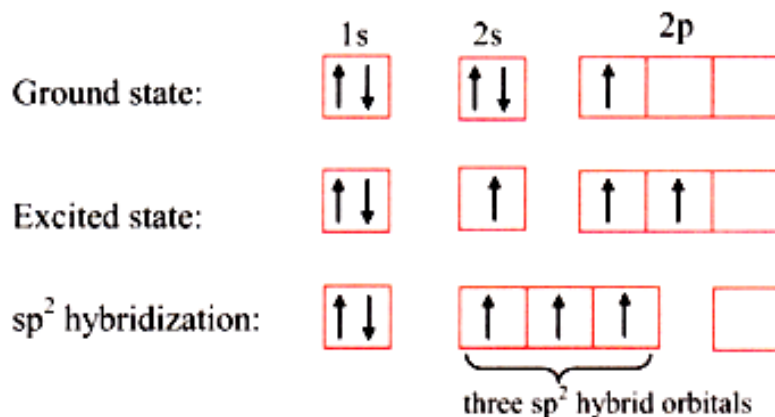
sp^2 hybridization

In this type, one s and two p orbitals (slightly different energy) intermix to form three sp^2 hybrid orbitals. Examples BCl_3 and C_2H_4



1. Boron dichloride (BCl_3):

- ✓ The electronic configuration of B is: $1s^2 2s^2 2p^1$
- ✓ Since, there is one unpaired electron. In this state B can make only single bond.
- ✓ For the formation of BCl_3 , three unpaired electrons are required.
- ✓ One electron is promoted from 2s to 2p orbital by absorbing energy.
- ✓ The electronic configuration in excited state becomes as $1s^2 2s^1 2p_x^1 2p_y^1 2p_z$

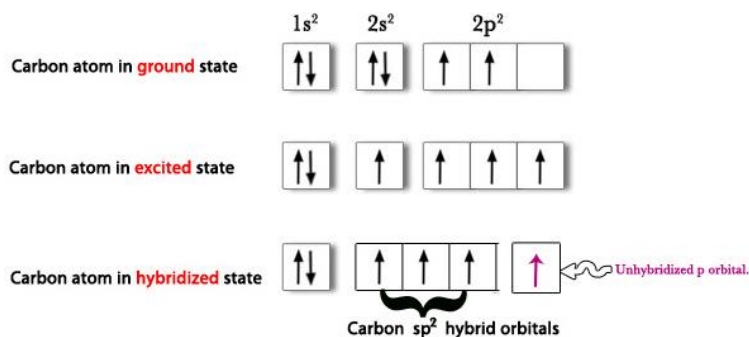


- ✓ In excited state there are three unpaired electrons.
- ✓ One $2s$ and two $2p$ atomic orbitals are intermixed to form three sp^2 hybrid orbitals with same energy and same shape.
- ✓ These three half-filled sp^2 hybrid orbitals are orientated in trigonal planer symmetry around the central atom.
- ✓ The electronic configuration of Cl is $1s^2 2s^2 2p^6 3s^2 3p_x^2 3p_y^2 3p_z^1$.
- ✓ It has one unpaired electron.
- ✓ This $2p$ atomic orbital of F overlap with sp^2 hybrid orbital of B to form trigonal planer geometry with 120° bond angle.



2.Ethylene (C_2H_4):

- ✓ The electronic configuration of C is $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$.
- ✓ It has only two unpaired electrons.
- ✓ On excitation, one electron from $2s$ atomic orbital is promoted to $2p$ orbital.
- ✓ Now four half-filled atomic orbitals are obtained for each carbon
- ✓ One $2s$ and two $2p$ atomic orbitals are intermixed to form three sp^2 hybrid orbitals.
- ✓ One half filled un-hybridized $2p$ orbitals is also present on each carbon perpendicular to the plane of sp^2 hybrid orbitals.

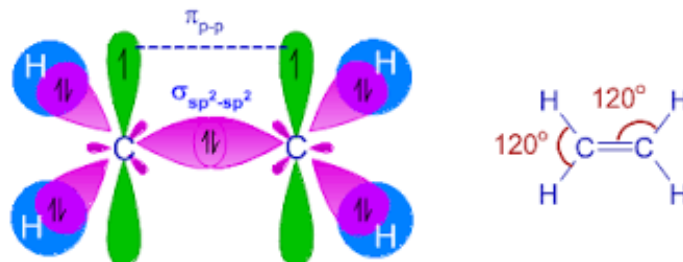


- ✓ Both carbons form the σ -bond with each other by sp^2-sp^2 head on overlap.

Ch#3: Chemical Bonding

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- ✓ Each carbon also forms the σ -bond with 1s atomic orbital of H.
- ✓ A π -bond is formed by the parallel overlap of un-hybridized half-filled 2p_z.
- ✓ So, double bond is formed between two carbon atoms (σ -bond & π -bond).
- ✓ Ethylene molecule is planer with HCH and HCC bond angle equal to 120°
- ✓ All the atoms are present in one plane.

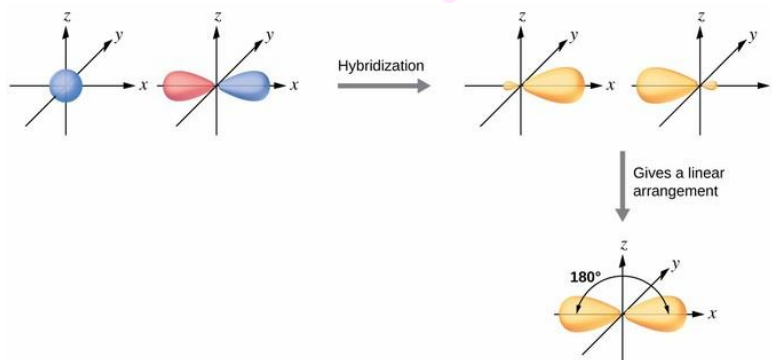
**sp hybridization:**

“The intermixing of one s and p atomic orbitals having different energy and different shape to form two sp hybrid orbitals having same energy and same shape is called sp hybridization.”

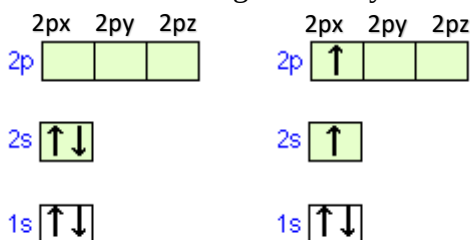
An sp hybrid orbital has 50%-s and 50%- p character.

sp hybrid orbital only form sigma bind with other atomic orbital.

Example Acetylene and BeCl₂

**Beryllium chloride (BeCl₂):**

- ✓ The electronic configuration of Be in ground state is $1s^2 2s^2$.
- ✓ There is no unpaired electron in the valence shell.
- ✓ On excitation one electron from 2s is promoted to 2p empty orbitals and electronic configuration become as $1s^2 2s^1 2p_x^1 2p_y^0 2p_z^0$.
- ✓ So, Be undergo sp hybridization by mixing of a 2s and one 2p atomic orbital.
- ✓ Two sp hybrid orbitals are formed which are arranged linearly around the Be.



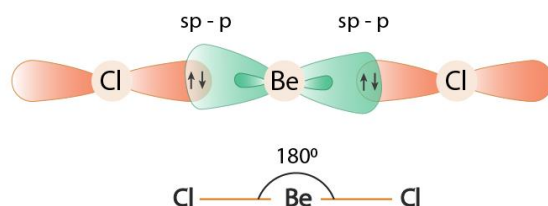
✓ Be atom in ground state Be atom in excited state

- ✓ The electronic configuration of Cl is $1s^2 2s^2 2p^6 3s^2 3p_x^2 3p_y^2 3p_z^1$.
- ✓ It has one half filled 3p_z orbital.

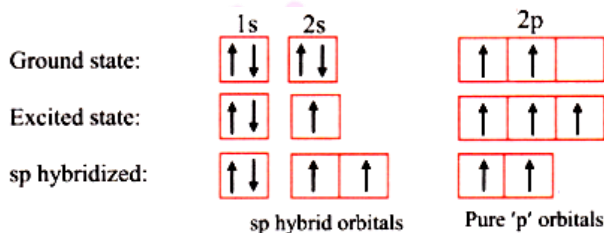
Ch#3: Chemical Bonding

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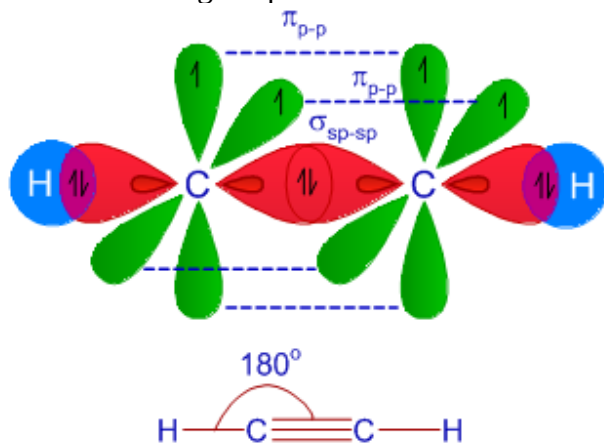
- ✓ The unpaired half filled 3p atomic orbital of each Cl overlap with sp hybrid orbital of Be to form the a sigma bond.
- ✓ The shape of BeCl_2 is linear with 180° bond angle.

2. Acetylene or Ethyne (C_2H_2):

- ✓ The electronic configuration of C is $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^0$.
- ✓ It has only two unpaired electrons.
- ✓ On excitation, one electron from 2s atomic orbital is promoted to 2p orbital.
- ✓ Now four half filled atomic orbitals are obtained for each carbon
- ✓ One 2s and one 2p atomic orbitals are intermixed to form two sp hybrid orbitals.
- ✓ Two half filled un-hybridized 2p orbitals are also present on each carbon perpendicular to the plane of sp hybrid orbitals.



- ✓ Both carbons form the σ -bond with each other by sp-sp head on overlap.
- ✓ Each carbon also form the σ -bond by sp-s overlap with 1s atomic orbital of H.
- ✓ However, two un-hybridized 2p_x and 2p_y orbitals are also present perpendicular to the plane of sp hybrid orbitals for each C.
- ✓ Two π -bonds are formed by the parallel overlap of un-hybridized half filled 2p_y and 2p_z atomic orbitals.
- ✓ So, triple bond is formed between two carbon atoms (one σ -bond & two π -bond).
- ✓ Acetylene molecule is linear with bond angle equal to 180° .



Co-ordinate covalent Bond (Dative bond)

"A bond between two atoms in which bonding electron pairs (share electron pair) is donated by one of the bonded atoms is called dative bond or co-ordinate covalent bond."

Atom that donates the electron has a lone pair of electrons and the other atom, which accepts the electron pair, has an unfilled empty orbital and is called an electron deficient atom or molecule.

Example

NH_3 and BF_3

Formation of O_3 (Ozone)

Formation of hydronium ion (H_3O^+)

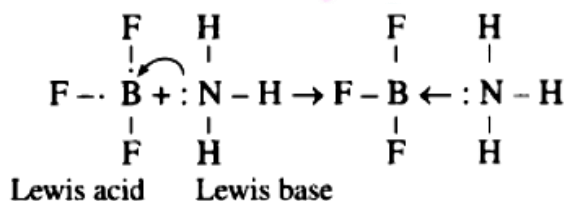
1. Bonding between NH_3 and BF_3 :

The electronic configuration of N and B is as follows



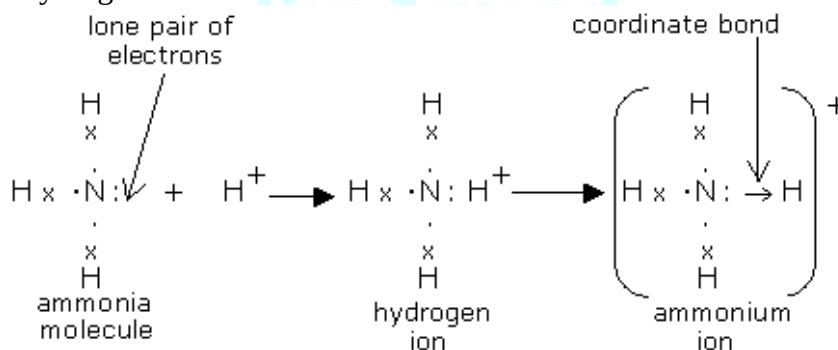
After BF_3 formation B has still an incomplete octet.

This octet is completed by the electron pair donated by Ammonia and a coordinate covalent bond is formed between them.



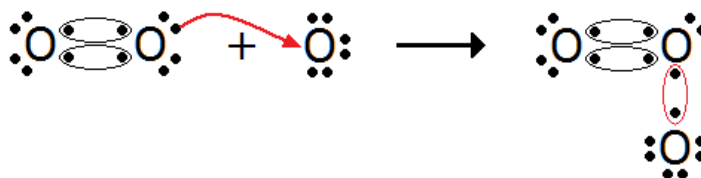
2. Formation of hydronium ion (H_3O^+):

During the formation of hydronium ion, water containing a lone pair of electrons donates a pair of electrons to the electron-deficient hydrogen atom and forms the dative covalent bond.



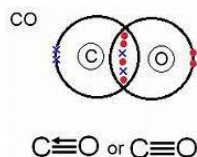
Formation of O_3 (Ozone):

In O_3 , central oxygen donates an electron pair to the other oxygen and forms a dative



4. Carbon monoxide (CO):

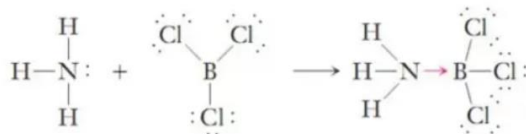
In the formation of CO, oxygen atom donates an electron pair to the Carbon and forms a coordinate covalent bond.



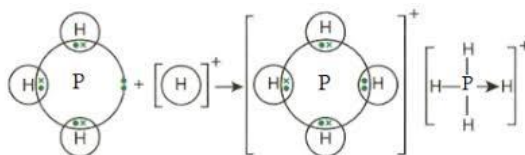
Concept Assessment Exercise 3.3

1. Dot and cross to show the coordinate covalent bond:

BCl_3 and NH_3 :



PH_3 and H^+ :



Molecular Orbital Theory (MOT)

According to this theory:

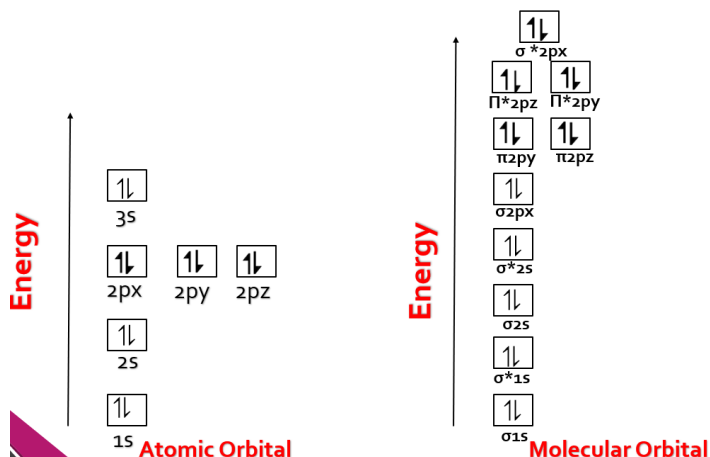
“Linear combination of atomic orbitals forms new orbitals called molecular orbitals which are the characteristics of whole molecule.”

Proposed by Hund and Mullikan in 1932.

Molecular approach considers the whole molecule as a single unit.

Postulates:

- ✓ Two atomic orbitals after overlapping form two molecular Orbitals (BMO and ABMO) which are differ in energy.
- ✓ Out of these, antibonding molecular orbitals always have higher energy than the parent orbitals.
- ✓ Molecular orbitals surround two or more nuclei of bonded atoms.
- ✓ BMO is symmetrical about the molecular axis while ABMO not.
- ✓ The postulates of Molecular Orbital Theory are:
- ✓ The total number of molecular orbitals formed is equal to the total number of atomic orbitals.
- ✓ The electrons in the molecular orbital are filled in the increasing order of orbital energy (from orbital having lower energy to orbital having higher energy). This filling is according to the Auf Bau principle, Pauli exclusion principle and Hund's Rule.



Difference between paramagnetic and Diamagnetic

Paramagnetic	Diamagnetic
Molecules having unpaired electrons in the molecular orbitals are paramagnetic in nature and can be attracted by magnetic field.	Molecules having no unpaired electrons in the molecular orbitals are diamagnetic in nature and cannot be attracted by magnetic field. They repel magnetic field.

Difference between BMOs and ABMOs

Bonding molecular Orbital (BMO)	Antibonding molecular Orbital (ABMO)
It is formed by addition overlap of atomic orbital	It is formed by subtraction overlap of atomic orbital
The energy of the bonding orbital is lower than that of atomic orbitals from which they are formed	The energy of ABMO is lower than that of atomic orbitals from which they are formed.
Bonding orbitals has high electron density in the region between the two nuclei which is responsible for the stability of bond.	It has zero electron density in the region between the two nuclei and concentration of electron density is on the opposite side.
Each electron in the BMO contribute the attraction between the two nuclei.	Each electron in the ABMO contribute the repulsion between the two nuclei.
They are designed as π and σ BMO	They are designed as π^* and σ^* ABMO

Difference between head on and sideways overlapping

Head on, linear overlapping or linear combination	Sideways or parallel Overlapping
Head on or linear overlapping takes by the combination of s-s, s-p and px-px orbitals.	Sideways or parallel overlapping of atomic orbitals takes place between py-py and pz-pz orbitals.
When "s" atomic orbitals overlap than σ_s and σ^*_s molecular orbitals are formed. σ_s is lower in energy and σ^*_s is higher in energy.	When two p orbitals py or pz overlap with each other than π_{py} , π_{pz} (lower in energy) and π^*_{py} , π^*_{pz} (higher in energy are formed.)

Energy for p orbitals:

When three 2p (px,py,pz) atomic orbitals of one atom overlap with three 2p atomic orbitals of other then six molecular orbitals are formed. Three are bonding molecular orbitals (σ_{2px} , π_{2py} , π_{2pz}) and three antibonding molecular orbitals (σ^*_{2px} , π^*_{2py} , π^*_{2pz}) are formed. The order of energy is;

$$\sigma_{2px} < \pi_{2py} = \pi_{2pz} < \pi^*_{2py} = \pi^*_{2pz} < \sigma^*_{2px}$$

Bond Order:

Total number of bonds when two atoms of atomic orbitals overlap with each other.

It is half of the difference between number of bonding electrons and number of antibonding electrons.

Bond Order: $\frac{\text{No. of electrons in BMOs} - \text{No. of electrons in ABMOs}}{2}$

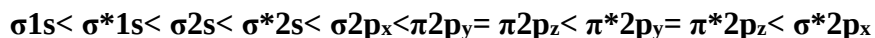
If the bond order is zero then molecule, then it does not exist.

A positive value shows the molecule is stable.

Relative energy of Molecular Orbitals:

From spectroscopic measurements form molecular orbitals formed from 2s and 2p atomic orbitals.

A) Following Increasing order of energy shows the molecular orbitals of diatomic molecule such O₂, F₂ and their ions.



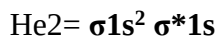
The diatomic molecules such as H₂, He₂, B₂, C₂, N₂ (lighter molecules) show slightly different energy order.



1. Hydrogen molecule (H₂)

Electronic configuration of H=1s¹

Total number of electrons in MOs=2

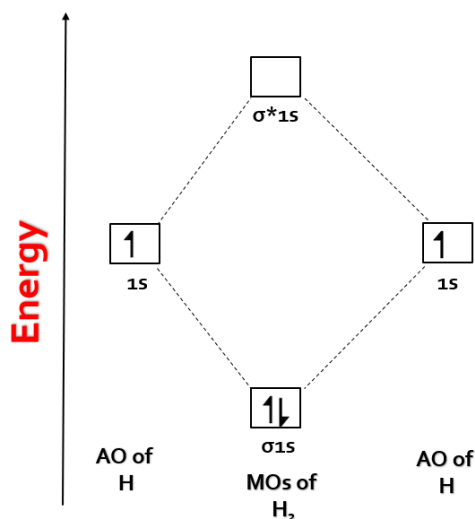


$$\text{Bond order} = \frac{2-0}{2} = 1$$

As bond order is 1. So, a stable single bond is existing between two hydrogen atoms.

Magnetic Character:

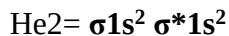
As there are no unpaired electrons in molecular orbital of H₂. It is diamagnetic.



2. He molecule (He₂)

Electronic configuration of He=1s²

Total number of electrons in MOs=4

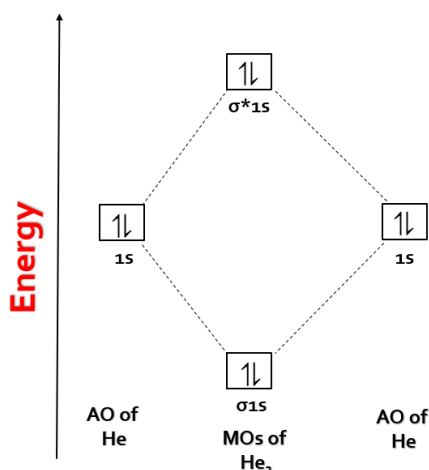


$$\text{Bond order} = \frac{2-2}{2} = 0$$

As bond order is zero. So, He₂ does not exist. He is monoatomic molecule.

Magnetic Character:

As there are no unpaired electrons in molecular orbital of He₂. It is diamagnetic.



3. Nitrogen Molecule: N_2

Electronic configuration = $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

Total number of electrons = 14

Total unpaired electrons = 6

Bond energy = 946 KJ/mol

Energetic order of MOs:

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_y^2 = \pi 2p_z^2 < 2\sigma p_x^2 < \pi^* 2p_y = \pi^* 2p_z < \sigma^* 2p_x$

Bond Order:

$$\text{Bond Order} = \frac{\text{No. of electrons in BMOs} - \text{No. of electrons in ABMOs}}{2}$$

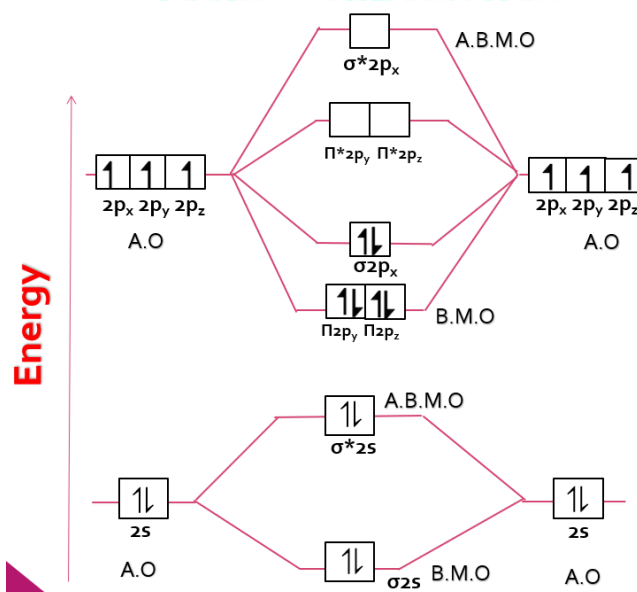
$$= \frac{6 - 0}{2}$$

Bond order = 3

So, triple bond exists between two nitrogen atoms.

Diamagnetic behavior:

As there are no unpaired electrons in the molecular orbitals. Therefore, Nitrogen molecule shows diamagnetic behavior.



4. Oxygen Molecule: O₂Electronic configuration = $1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$

Total number of electrons = 16

Total unpaired electrons = 2

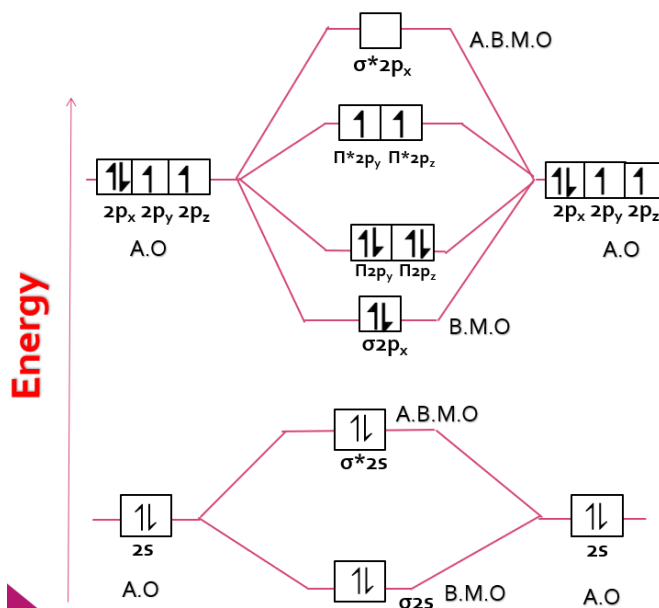
Energetic order of MOs: $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < 2\sigma p_x^2 < \pi 2p_y^2 = \pi 2p_z^2 < \pi^* 2p_y^1 = \pi^* 2p_z^1 < \sigma^* 2p_x$ **Bond Order:**

$$\text{Bond Order} = \frac{\text{No. of electrons in BMOs} - \text{No. of electrons in ABMOs}}{2}$$

$$= \frac{6 - 2}{2}$$

Bond order = 2

So, a stable double bond exists between two oxygen atoms.

**Paramagnetic behavior of O₂:**

Two unpaired electrons reside in the degenerate antibonding molecular orbitals $\pi^* 2p_y^1$ and $\pi^* 2p_z^1$. They electrons not only go around the atoms in their orbitals but also spin which creates the magnetic field. They spin in the same direction which increase the magnetic field effect.

Due to these unpaired electrons O₂ is found to be paramagnetic.This paramagnetic effect of O₂ was not explained by the VBT.Bond length: $1.21 \text{ \AA}$

Bond Energy = 498 KJ/mol

5. Fluorine Molecule: F₂✓ Electronic configuration = $1s^2 2s^2 2p_x^2 2p_y^2 2p_z^1$

✓ Total number of electrons = 20

✓ Total unpaired electrons = 1

Energetic order of MOs: $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < 2\sigma p_x^2 < \pi 2p_y^2 = \pi 2p_z^2 < \pi^* 2p_y^2 = \pi^* 2p_z^2 < \sigma^* 2p_x$ **Bond Order:**

$$\text{Bond Order} = \frac{\text{No. of electrons in BMOs} - \text{No. of electrons in ABMOs}}{2}$$

$$= \frac{6 - 4}{2}$$

Ch#3: Chemical Bonding

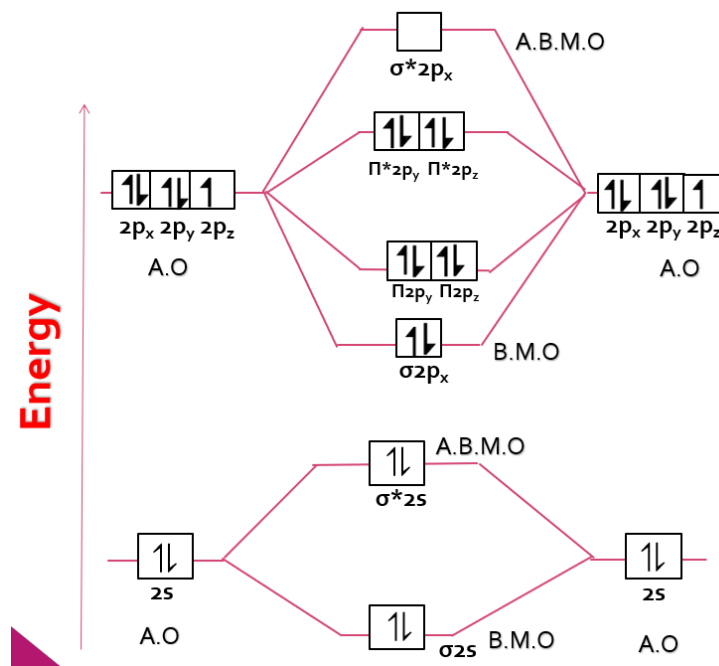
By: Sir Amjad (Lecturer Chemistry)

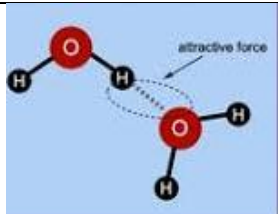
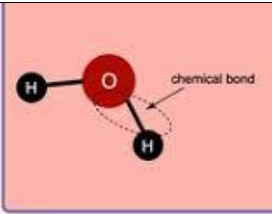
Bond order=1

So, a stable single bond exists between two fluorine atoms.

Diamagnetic Behavior:F₂ is diamagnetic in nature due to the absence of unpaired electrons in molecular orbitals.**Bond energy:** 159KJ/mol

Bond length: 1.43 Å

**Intermolecular forces (Forces between the molecules)****Difference between intermolecular and intramolecular forces**

The forces which are present between the molecules are called intermolecular forces.	The forces which are present within a molecule are called intramolecular forces.
Weak forces	Strong forces
These are attractive forces	These are chemical bonds
	

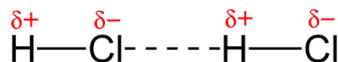
Types of intermolecular forces

These are three types of intermolecular forces

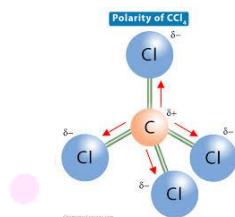
- ✓ Permanent Dipole-dipole forces
- ✓ Instantaneous dipole induced dipole forces
- ✓ Hydrogen Bindings

1. Permanent Dipole-dipole forces:

- ✓ These are the weak forces of attractions present between two polar molecules.
- ✓ Due to electronegativity difference one atom with high electronegativity gets partial positive charge and other atom with less electronegativity gets partial negative charge.
- ✓ The greater the polarity of a bond or molecule, the stronger the permanent dipole-dipole interactions.
- ✓ For example, in HCl, H gets δ^+ and Cl gets δ^- charge. So, it has overall dipole moment.
- ✓ The δ^+ end of H attracts the δ^- end of Cl and vice versa. These attractions are dipole-dipole attractions.



- ✓ But in some molecules like CCl_4 , the individual dipoles cancel out each other, so, CCl_4 is a non-polar molecule.



- ✓ Stronger the dipole-dipole forces, higher will be the value of melting point, boiling point, heat vaporization and heat of sublimation of a compound.

Instantaneous dipole induced dipole forces or London Dispersion forces or temporary dipole forces

“The attractive forces present between temporary dipole of one molecule and induced dipole in other molecules are called instantaneous dipole induced dipole forces.”

Explanation:

Present in all types of molecules (polar and non-polar).

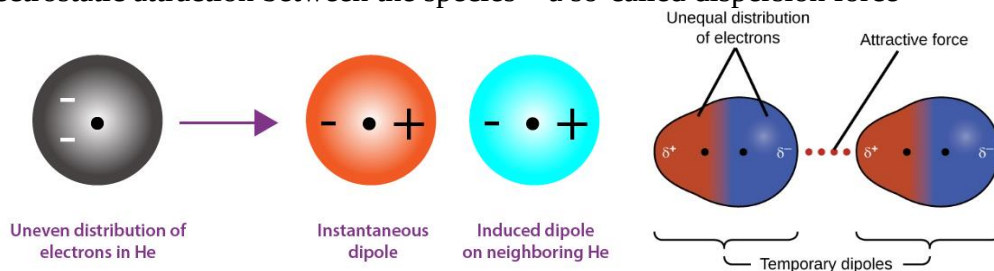
Polar molecules have only London dispersion forces (Noble gases, Halogens, CH_4 etc)

A German physicist Fritz London

He is non-polar and can be liquefied.

Because the electrons of an atom or molecule are in constant motion at any moment in time, an atom or molecule can develop a temporary, **instantaneous dipole** if its electrons are distributed asymmetrically.

The presence of this dipole can, in turn, distort the electrons of a neighboring atom or molecule, producing an **induced dipole**. These two rapidly fluctuating, temporary dipoles thus result in a relatively weak electrostatic attraction between the species—a so-called dispersion force

**Strength of dipole-dipole and instantaneous dipole induced dipole forces (Van der Waals forces)**

Strength depends upon the following factors.

- ✓ Molecules size---(Bigger the molecule)
 - ✓ Atomic Size ---Bigger Size
 - ✓ Molecular mass---(higher relative molecular mass)
 - ✓ Number of electrons in a molecule. -----(more electrons)
- = More van der Waals forces

Table 3.5: Boiling point of first four members of alkane

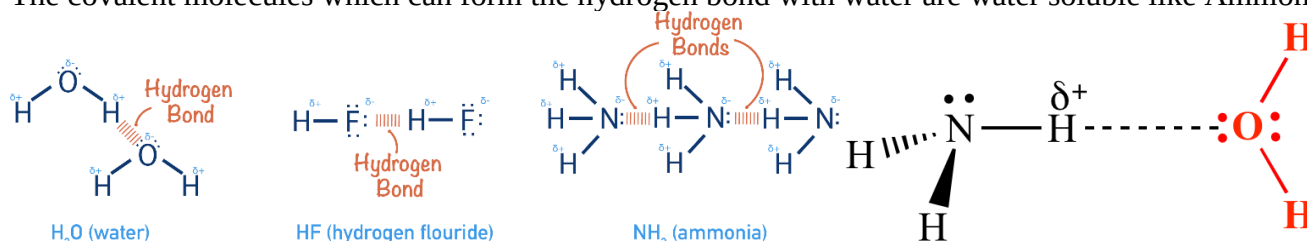
Name	Molecular formula	Boiling point (°C)
Methane	CH ₄	-164
Ethane	C ₂ H ₆	-89
Propane	C ₃ H ₈	-42
Butane	C ₄ H ₁₀	-1

Hydrogen Bonding:

The forces of interaction between the highly electron deficient hydrogen and lone pair of electron of highly electronegative atom such as O, N and F are called hydrogen bonding.

These are the strongest force of attractions between the molecules.

The covalent molecules which can form the hydrogen bond with water are water soluble like Ammonia.

**Peculiar Behavior of Hydrogen Bonding:****Why Ice floats on water?**

Most of the solid are denser than liquid. Because in solid state molecules are more closely packed. Because this is not true for water due to following reason.

✓ **Crystal structure**

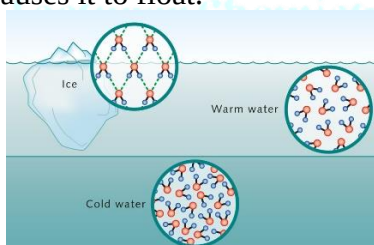
When water freezes, its molecules form a crystalline structure held together by hydrogen bonds. The orientation of these bonds pushes the molecules farther apart, which lowers the density of the ice.

✓ **Expansion**

As water freezes, it expands, so ice has more volume than water, but less density.

✓ **Density**

Density is directly proportional to mass and inversely proportional to volume. Ice has a density of 0.931 grams per cubic centimeter, while water has a density of 1 gram per cubic centimeter. Because ice is lighter than water, the water displaces it and causes it to float.

**Comparison of strength of intermolecular forces:**

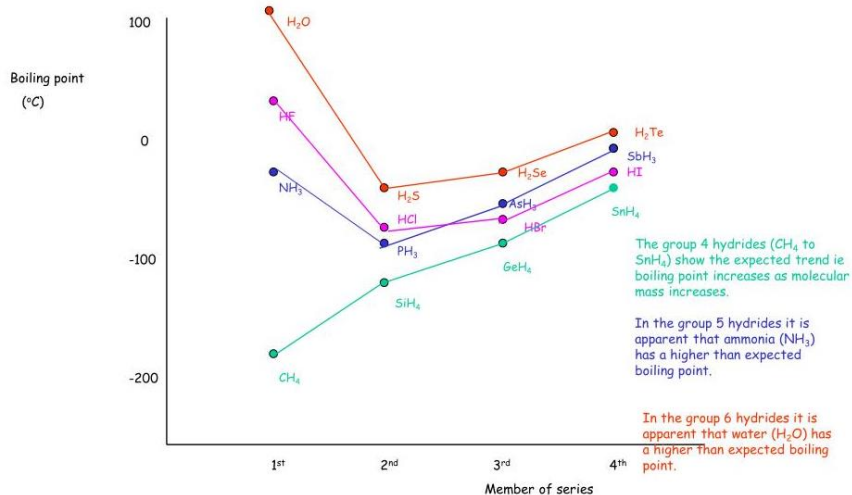
The general order of strength is

H-bonding > permanent dipole-dipole > Instantaneous dipole-induced dipole

Boiling points:

Molecule	CH ₄	HCl	H ₂ O
Boiling point (°C)	-162	-85	100
Intermolecular forces	Instantaneous dipole-induced dipole	permanent dipole-dipole	H-bonding

Boiling points of hydrides of groups 4, 5, 6 and 7.



The higher than expected boiling points are due to hydrogen bonding.

Comparison of strength of ionic, covalent, metallic bondings and intermolecular forces:

Ionic Bond: These are the electrostatic forces of attraction between the positive and negative charges.

Covalent Bond: This is a type of bond formed by the sharing of electrons between the two atoms. The electrons are held in the region between the two nuclei of bonded atoms.

Metallic bond: This is type of bond between the two metals where electrons are shared and free to move throughout the material. This creates a sea of delocalized electrons that hold the metals together.

Intermolecular forces: These are the interactions between the molecules. These are responsible to hold the molecules together.

The general trend of strength is;

Ionic Bond > Covalent Bond > Intermolecular forces

Metallic Bonding

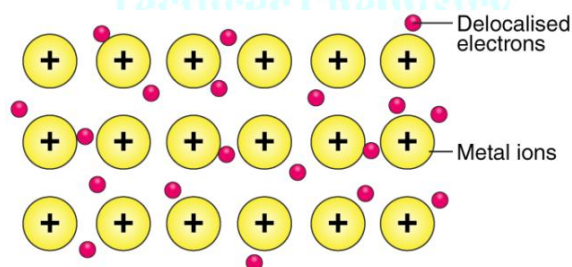


Table 3.7: Following table represents mp/bp of ionic, simple covalent, giant covalent and metallic substances.

Substance	Bonding / structure	Melting point/°C	Boiling Point/°C
Sodium Chloride	Ionic (Electrostatic forces)	801	1465
Diamond	Giant Covalent (Covalent bond)	4000	4830
Iron	Metallic (electrostatic forces)	1538	2862
Methane	Simple Covalent (Intermolecular forces)	-182	-161.6

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